# Muhammad Bilal

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## SUMMARY

Fulbright Scholar pursuing an MS in Computer Vision at Northeastern University and the Gordon Institute of Engineering Leadership. Software engineer and technical lead who ships end-to-end systems: computer vision on live video, neural inverse kinematics for robotic arm control, multilingual RAG platforms, and full-stack microservice products for enterprise clients. 4.0/4.0 CGPA with the President's Gold Medal, and a peer-reviewed ICED 2025 publication.

## EDUCATION

**Northeastern University, Boston, MA** Sep 2026 – 2028 (expected)

*MS in Electrical and Computer Engineering (Computer Vision)*

- **Fulbright Scholar**, Fulbright Foreign Student Program (U.S. Department of State)
- **Gordon Institute of Engineering Leadership**, Graduate Certificate in Engineering Leadership

**National University of Sciences and Technology (NUST), Pakistan** Graduated Jun 2025

*BS in Software Engineering | CGPA 4.0/4.0 | President's Gold Medal (1st in class)*

## EXPERIENCE

**Technical Team Lead & Developer (Freelance), MHTechFusion** Dec 2024 – Aug 2026

*Technical Team Lead* Dec 2025 – Aug 2026

- **Splatoon AI:** Directed an AI research project that analyzed live game video feeds with computer vision to drive aimbot, movement, and AI-teammate models; resolved edge cases where real-world lighting changes and sensor noise made inference unstable.
- **Cradle AI:** Led development of an enterprise-grade AI governance platform with a custom multilingual RAG engine and dedicated web interface.
- **RTTA (Real-Time Translation App):** Led the architecture of an iOS application performing real-time, low-latency audio translation during live calls.

*Software Developer* Dec 2024 – Nov 2025

- Built full-stack web applications and secure, data-driven dashboards with React, Node.js, PostgreSQL, and Supabase.

**Software Engineer, GoSaaS Inc.** Jun 2025 – Aug 2026

*Client: SharkNinja*

- Supported and developed system enhancements for SharkNinja's Finance and Inventory modules.
- Designed and maintained enterprise workflows and customizations using Oracle OIC, ATP, APEX, and VBCS.

*Product Team* Jun 2025 – Feb 2026

- Engineered GoCAD, a full-stack application that streamlines workflows between CAD tools and Oracle PLM.
- Built cross-platform ElectronJS desktop interfaces on a microservices architecture (React, Node.js, PostgreSQL).
- Developed centralized, scalable logging microservices with Docker containerization.

**Research Assistant, ROMI Lab, NUST** Oct 2024 – Jun 2025

- Designed backend pipelines and web interfaces to visualize real-time system outputs and optimize scalable design.

**MITACS Globalink Research Intern, ÉTS Montréal, Canada** Jun 2024 – Sep 2024

- Built a full-stack web platform (React, Node.js, Neo4j) for dynamic, graph-based data modeling.
- Published peer-reviewed research at ICED 2025 (International Conference on Engineering Design) based on the system's architecture and platform design.

## SKILLS

**AI & Computer Vision:** Python, TensorFlow/Keras, OpenCV, YOLOv8, RAG engines, LLM integration
**Languages:** Python, JavaScript (ES6+)
**Full-Stack:** React, Node.js, Express.js, ElectronJS, REST APIs, microservices, HTML5, CSS3
**Data:** SQL, PostgreSQL, MongoDB, Supabase, Neo4j
**Cloud & Enterprise:** Docker, AWS, CI/CD, Git, Oracle (OIC, ATP, APEX, VBCS)

## HONORS & LEADERSHIP

- **Fulbright Scholarship** for graduate study in the U.S. (Northeastern University, 2026)
- **President's Gold Medal**, NUST (1st position, BS Software Engineering)
- **MITACS Globalink Research Grant** (fully funded research internship, Canada)
- **Prime Minister's Laptop Award** (top 20 students nationwide)
- **Leadership:** Final Year Project Team Lead (2024–25); Deputy Director, NUST Archery Club; Executive, NUST Leadership Society